

HeliOps - Installation, Dependencies and External DCS Modifications

Complete installation and version-lock guide for the current HeliOps configuration

This guide replaces the earlier External DCS Modifications document. It covers the HeliOps project tree, bundled MOOSE version, external SRS and DCS-gRPC dependencies, the two DCS files modified outside HeliOps, bridge requirements, load order and post-install verification.

1. Tested configuration

Component	Tested version	Role / location
DCS World	2.9.29.27278	MT build, Windows
MOOSE	commit c48a7d4400161ef6f90967878f2b125ebe 7f52ab	Bundled: lib\Moose.lua
MOOSE FLIGHTCONTROL	0.7.7	Used for native ATC/player callbacks
MOOSE MSRS	0.3.6	TTS/radio layer
DCS-SimpleRadio-Standalone	2.4.0.0	External installation
DCS-gRPC	0.8.1	External installation / gRPC backend
HeliOps RadioBridge	.NET 8 target	HeliOps bridge\...

Version policy

Treat a change to MOOSE, SRS, DCS-gRPC or DCS itself as a dependency upgrade. Keep lib/Moose.lua version-locked to the HeliOps version and retest the radio/ATC chain after upgrades.

2. Recommended HeliOps repository layout

```
HeliOps\
framework\
  HeliOps.lua
  HeliOps_Airports.lua
  HeliOps_FlightControl.lua
  HeliOps_RadioRouter.lua
  HeliOps_RadioDevices.lua
  ...
scripts\
export\
  HeliOps_RadioExport.lua
bridge\
lib\
  Moose.lua
dependencies\
  README.md
  versions.md
docs\
missions\
  README.md
```

Current runtime root:

```
C:\Users\<USER>\Saved Games\DCS\Missions\HeliOps\
```

MOOSE remains inside lib/ because HeliOps loads and tests against a specific bundled version. SRS and DCS-gRPC are external dependencies and should not be copied wholesale into lib/.

3. Bundled dependency: MOOSE

- Keep the tested MOOSE file in HeliOps\lib\Moose.lua.
- Do not modify MOOSE locally unless the change is intentionally maintained as a fork.
- When replacing Moose.lua with a newer upstream version, retest FLIGHTCONTROL, FLIGHTGROUP, MSRS/gRPC and player ATC callbacks.

Current status

The HeliOps work described here did not modify MOOSE source code. The bundled copy exists for deterministic loading and version compatibility.

4. External dependency: DCS-SimpleRadio-Standalone (SRS)

Install SRS separately. HeliOps does not bundle SRS.

```
C:\Program Files\DCS-SimpleRadio-Standalone\
```

Current tested version: 2.4.0.0. HeliOps/MSRS uses SRS ExternalAudio on port 5002.

```
SRS ExternalAudio: 127.0.0.1:5002
```

- Install/start SRS normally.
- Keep the SRS export integration in Saved Games\DCS\Scripts\Export.lua.
- Do not replace the SRS Export.lua block with HeliOps code; HeliOps must coexist with it.
- For HeliOps FlightControl, tower and pilot MSRS instances use the gRPC backend while SRS supplies the radio audio path.

5. External dependency: DCS-gRPC

Install DCS-gRPC separately under the normal Saved Games DCS structure. Current tested version: 0.8.1.

```
C:\Users\<USER>\Saved Games\DCS\Scripts\DCS-gRPC\
```

The current HeliOps integration uses:

```
gRPC server: 127.0.0.1:50051
```

```
SRS target : 127.0.0.1:5002
```

```
TTS provider: Windows (current configuration)
```

- DCS-gRPC must autostart with the simulation or otherwise be running before HeliOps uses the bridge/TTS path.
- The mission loader must be called from DCS World\Scripts\MissionScripting.lua before sanitization.
- Standard sanitization remains enabled; HeliOps does not require os, io or lfs to remain exposed to the mission environment.

6. External DCS modification #1: MissionScripting.lua

File:

```
C:\Program Files\Eagle Dynamics\DCS World\Scripts\MissionScripting.lua
```

Immediately after the ScriptingSystem loader, add the DCS-gRPC mission loader:

```
dofile('Scripts/ScriptingSystem.lua')
dofile(lfs.writedir()..[[Scripts\DCS-gRPC\grpc-mission.lua]])

-- Sanitize Mission Scripting environment
```

Important

Do not remove or comment out the standard sanitization block. DCS Update/Repair may overwrite MissionScripting.lua, so verify this line after DCS updates.

Verify that the loader target exists:

```
C:\Users\<USER>\Saved Games\DCS\Scripts\DCS-gRPC\grpc-mission.lua
```

7. External DCS modification #2: Export.lua

File:

```
C:\Users\<USER>\Saved Games\DCS\Scripts\Export.lua
```

Load the HeliOps export file using the current HeliOps runtime root:

```
dofile(
  lfs.writedir()
  .. [[Missions\HeliOps\export\HeliOps_RadioExport.lua]]
)
```

Path correction

Older documentation used Saved Games\DCS\HeliOps\... . The current project root is Saved Games\DCS\Missions\HeliOps\... . Use the Missions\HeliOps path above.

8. Export.lua coexistence and load order

Export.lua is shared by several DCS integrations. Preserve existing SimShaker, SRS and VAICOM blocks. The current safe arrangement is to load HeliOps after the existing exporters so it can chain the previously registered callbacks.

```

-- Existing exporters first, for example:
-- SimShaker
-- DCS-SRS
-- VAICOM

-- HeliOps LAST
dofile(
    lfs.writedir()
    .. [[Missions\HeliOps\export\HeliOps_RadioExport.lua]]
)

```

HeliOps_RadioExport.lua must use LuaExportAfterNextFrame and chain the previous callback first. It must not participate in or replace LuaExportActivityNextEvent. This avoids the SRS disconnect/callback collision encountered during development.

9. HeliOps RadioBridge

The resident bridge transfers cockpit radio state from the Export environment to the mission/gRPC side and accepts VoiceAttack command requests.

Current executable build location:

```
C:\Users\<USER>\Saved Games\DCS\Missions\HeliOps\bridge\bin\Release\net8.0\win-x64\publish\HeliOpsBridge.exe
```

Bridge endpoints:

```

Radio UDP : 127.0.0.1:9088
Command UDP : 127.0.0.1:9089
DCS-gRPC : http://127.0.0.1:50051

```

- Run one resident HeliOpsBridge.exe instance with no arguments.
- VoiceAttack launches the same executable with station and command arguments, for example: AKROTIRI RADIO_CHECK or AKROTIRI INBOUND.
- If rebuilding the bridge from source, use the .NET 8 SDK because the project targets .NET 8.
- After changing Program.cs command definitions, rebuild/publish and restart the resident bridge.

10. HeliOps radio and FlightControl requirements

Current HeliOps FlightControl explicitly selects the MOOSE MSRS gRPC backend for both tower and pilot instances. This is the configuration that removed the multi-second simulator freeze seen with the external executable TTS path.

```

MSRS.SetDefaultBackendGRPC()
fc.msrsTower:SetBackend(MSRS.Backend.GRPC)
fc.msrsPilot:SetBackend(MSRS.Backend.GRPC)

```

Current voice configuration:

```

Tower: female / en-GB
Pilot: male / en-US

```

OH-58D UHF radio state is read from cockpit device 30 (AN/ARC-164). HeliOps validates the spoken station against the actual tuned cockpit frequency before dispatching the native MOOSE command.

11. Startup and verification

After installation or an update, verify the complete chain in this order:

- DCS-gRPC starts and listens on 127.0.0.1:50051.
- SRS starts normally and remains connected.
- The resident HeliOpsBridge starts and reports UDP 9088/9089 plus the gRPC endpoint.
- HeliOps mission loads without SCRIPTING ERROR.
- HeliOps FlightControl log shows towerBackend=grpc and pilotBackend=grpc.
- Tune the OH-58D UHF radio to the correct airport frequency.
- Say "Akrotiri, radio check". Expected tower reply: "Scout, Akrotiri Tower, readability five."
- Test "Akrotiri, request taxi" and confirm the native MOOSE taxi callback responds.
- After takeoff, test "Akrotiri inbound" and confirm MOOSE recognizes Scout as airborne.

12. Current airport frequencies

Airport	Frequency	Modulation
Akrotiri	251.700 MHz	AM
Paphos	251.800 MHz	AM
Larnaca	251.850 MHz	AM

13. Post-update checklist

- MissionScripting.lua still contains the grpc-mission.lua loader directly after ScriptingSystem.lua.
- The normal sanitization block is still active.
- Export.lua still contains the Missions\HeliOps\export\HeliOps_RadioExport.lua loader.
- SimShaker/SRS/VAICOM exporters have not been removed or overwritten.
- HeliOps is still loaded last in the export chain.
- lib\Moose.lua is the intended version for this HeliOps revision.
- SRS and DCS-gRPC versions match the tested dependency record or have been deliberately upgraded and retested.
- RADIO_CHECK, TAXI, TAKEOFF and INBOUND all function after the update.

Repository recommendation

Store this guide under HeliOps\docs and keep dependencies\versions.md in sync with it whenever DCS, MOOSE, SRS, DCS-gRPC or the bridge target changes.